The Parser-Rescue Gap: Why Small-Data Fine-Tuning Inflates Accuracy on Structured Prediction

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Abstract

I compare QLoRA fine-tuning of Llama-3.1-8B against in-context learning (ICL) and chain-of-thought prompting (CoT) on a 192-example structured numerical reasoning task: FIFA World Cup match prediction from per-player statistics. Standard accuracy places QLoRA (n=192) at 61.7% on named inputs, seven points above ICL (49.2%). A follow-up analysis of output *internal consistency* shows this gap is largely an artifact of the evaluation convention. The fine-tuned model's `Prediction:` label and `Score:` line are generated nearly independently: on the in-distribution named split, when the two disagree, the score matches ground truth 6–14× more often than the text label for QLoRA at $n \geq 96$, while for ICL and CoT the ratio reverses. Under a coherence-required metric that credits only outputs in which the text label, score, and ground truth all agree, ICL and QLoRA n=192 are *tied* at 42.2% (±8.4 pp 95% Wilson CI), and the apparent fine-tuning advantage vanishes. Small-data fine-tuning learns output *format* with near-perfect fidelity but does not bind fields into a coherent commitment; the common "parse any reliable field" evaluation pattern rewards this decoupling in this setting. I argue that benchmarks over structured multi-field generative outputs should report coherence-required accuracy alongside primary metrics as a cheap check for this failure mode.

1 Introduction

Many real-world prediction tasks involve structured numerical data with scarce labeled examples. I use FIFA World Cup match prediction as a controlled testbed: each match provides dozens of per-player statistics, the outcome is unambiguous, and the dataset size is *inherently* small — 64 matches per tournament, held every four years. The original question of this project was straightforward: on $\sim$192 labeled examples, does QLoRA fine-tuning teach an 8B transformer to weight statistical features, or are prompting-based methods (ICL, CoT) competitive?

The comparison went as planned: QLoRA reaches 52.3% result accuracy on the 2022 eval set (7 points above the always-home-win baseline of 45.3%), ICL reaches 49.2%, and CoT 44.5%, confirming the proposal's hypothesis H1 (fine-tuning > ICL). But two observations redirected the analysis. ICL and CoT produce `home_win` on 82% and 99% of samples respectively against a 45% GT rate — they are not reasoning but collapsing to the home-advantage prior. And the fine-tuned `Prediction:` line and `Score:` line are generated nearly independently: QLoRA accuracy drops from 52.3% to 39.8% if I use only what the model writes in its `Prediction:` line. The standard eval convention of "score overrides text" is accidentally *rescuing* the fine-tuned model while *penalizing* the more coherent prompting alternatives.

Section 3.1 reports the numbers the proposal called for. Section 3.2 reports the methodological finding — that small-data fine-tuning teaches output format with near-perfect fidelity but fails to bind fields into a coherent commitment, and that standard metrics can reward this failure. Section 4 connects these findings to course material (L12, L17–20) and to the specification-gaming literature. My central claim is not that a single model and dataset resolve this question, but that the parser-rescue gap is a specific, diagnosable failure mode worth checking whenever small-data fine-tuning is evaluated with a "parse any reliable field" convention on multi-field structured outputs.

2 Task, Dataset, and Method

Task. Given two teams' aggregated player statistics — squad goal output, average rating, pass completion, top scorer output, defensive stats — the model predicts the match result ($\in$ {`home_win`, `draw`, `away_win`}) and final score. The expected output format is three lines: `Prediction: <label>`, `Score: <h>-<a>`, and `Reasoning: <sentence>`.

Dataset. 256 FIFA World Cup matches across 2010–2022 (64 per tournament), built from 100,000+ player-season entries via API-Football. For each match, both teams' starting XI are identified and each starter's league statistics from *prior seasons only* are aggregated into a team profile to prevent temporal leakage. I split by tournament: train on 2010+2014+2018 (192 matches) and eval on 2022 (64 matches). Each set has a *named* variant (real team names) and an *anonymized* variant (Team A / Team B), probing whether pre-training priors on team reputation interact with statistical reasoning (proposal H3).

QLoRA fine-tuning. Llama-3.1-8B-Instruct with 4-bit NF4 quantization (double quantization, `float16` compute dtype), LoRA rank 16 with α=32 and dropout 0.05, targeting the seven standard attention and MLP projections [3, 4]. Trained 3 epochs with effective batch size 16, cosine schedule, learning rate 2×10^{-4}, warmup 10%, weight decay 0.01, gradient checkpointing, on a single Colab T4. For the scaling ablation I train additional adapters at $n \in \{48, 96, 192\}$ using *nested* stratified prefixes (verified $s_{48} \subset s_{96} \subset s_{192}$) drawn from the

named training subset. The n=384 configuration reuses the existing `zanwenfu/football-llm-qlora` adapter, trained on 192 named + 192 anonymized examples.

ICL and CoT baselines. For ICL, base Llama-3.1-8B-Instruct is given 5 few-shot demonstrations from the named training set, stratified 2 home / 2 away / 1 draw with `seed=42`; anonymized eval samples receive anonymized versions of the same demonstrations. For CoT, the base model is given a 5-step reasoning scaffold (compare goal output; compare top scorer; compare defense; weigh conflicts; predict) with `max_new_tokens=1024` to ensure the reasoning reaches a final `Prediction:` line. No parameter updates in either condition.

Evaluation protocol. All conditions are evaluated on 128 samples (64 named + 64 anonymized) with temperature 0.1, top-p 0.9, and `repetition_penalty=1.1`. The parser extracts a *text label* from the `Prediction:` line, parses a *score label* from the `Score:` line, and when both are parseable uses the score (the standard eval_harness convention I later interrogate). All reported confidence intervals are Wilson 95% two-sided.

3 Results

3.1 What standard metrics suggest

Table 1 (score-acc column) reports standard score-based accuracy across QLoRA budgets and prompting baselines. Fine-tuned adapters at $n \geq 192$ appear to outperform both prompting baselines; n=192 reaches 61.7% overall (71.9% on named inputs). ICL and CoT hover near the 45.3% always-home-win baseline. The non-monotonic n=96 dip could reflect training-set class proportions at this budget or seed variance; I train a single seed per n and do not attempt to distinguish these without multi-seed runs. If analysis stopped here, the standard story would be: fine-tuning on structured data beats prompting, scaling helps up to n=192, and the drop at n=384 reflects a mixing effect from the anonymized augmentation.

The prediction-distribution analysis reveals a first anomaly, however. ICL predicts `home_win` on 82% of samples and CoT on 99% (Table 1, rightmost column), against a 45% GT rate. Prompting-based methods are not reasoning — they are collapsing to the home-advantage prior. CoT's elaborate reasoning text is in fact decoupled from its output label in a specific way: the argument cites stats for whichever team is statistically stronger, but the final label is `home_win` regardless. This collapse alone explains why prompting-based accuracy sits at baseline; with a nearly-constant prediction, result accuracy equals the GT home-win rate. QLoRA adapters at $n \geq 96$ produce more diverse predictions (home 21–52%, away 44–60%, draw 2–19%), breaking out of this collapse — but, as §3.2 shows, not in the way one would expect.

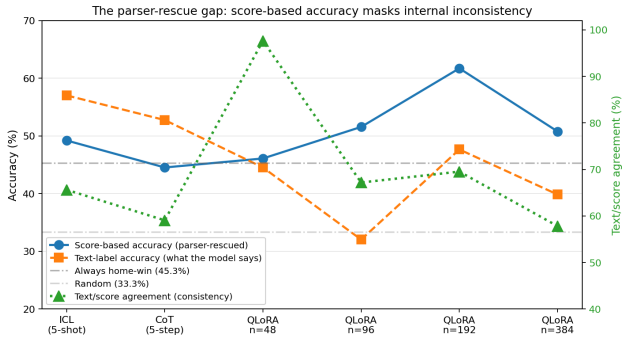

Figure 1: The parser-rescue gap. Solid blue: score-based accuracy. Dashed orange: text-label accuracy (what `Prediction:` actually says). Dotted green (right axis): text/score agreement. The n=48 spike is majority-class collapse, not coherence. For ICL/CoT the parser penalizes; for QLoRA at $n \geq 96$ the parser rescues.

Condition	Score acc	Text acc	Coherent acc (A)	P(home) pred
Always home-win	45.3%	—	—	100%
Random weighted	35.9%	—	—	40%
ICL (5-shot)	49.2% ±8.5	**57.0%** ±8.5	<u>42.2%</u> ±8.4	82%
CoT (5-step)	44.5% ±8.5	**52.3%** ±8.5	35.2% ±8.2	99%
QLoRA n=48	46.1% ±8.5	44.5% ±8.5	44.5%†	98%
QLoRA n=96	51.6% ±8.5	32.0% ±8.0	28.1% ±7.7	21%
QLoRA n=192	**61.7%** ±8.3	47.7% ±8.5	<u>42.2%</u> ±8.4	45%
QLoRA n=384	50.8% ±8.5	39.8% ±8.4	32.0% ±8.0	52%

Table 1: Three views of accuracy with 95% Wilson CI (n=128). **Bold**: best per column within fine-tuned/prompting group. <u>Underline</u>: ICL and QLoRA n=192 tied on coherent acc. † n=48 coherence is trivial (model emits `home_win` on 98% of inputs).

3.2 What internal consistency reveals

Three views of a single output. Each model response contains three fields: a `Prediction:` label, a `Score:` line, and a `Reasoning:` paragraph. The standard eval convention extracts a *text label* from `Prediction:`, parses a *score label* from `Score:`, and uses the score label when both exist. This convention was designed for rare inconsistencies — e.g. the model writes `home_win` with a `0-1` score — that a well-calibrated model should produce only occasionally. My data shows these "rare" inconsistencies are the dominant failure mode of small-data fine-tuning, making the 61.7% headline number substantially an artifact of the eval convention rather than of what the model writes.

Figure 1 shows the gap between score-based and text-label accuracy. It is large, systematic, and *directionally opposite* between fine-tuned and prompting-based conditions: text accuracy exceeds score accuracy by 7.8 pp for ICL and 8.3 pp for CoT (parser penalty), whereas score accuracy exceeds text accuracy by 11–20 pp for QLoRA at $n \geq 96$ (parser rescue). Table 1 consolidates this together with a coherence-required metric introduced below.

Condition	Disag.	Score=GT	Text=GT
ICL	20	4	**9**
CoT	26	6	**12**
QLoRA n=48	2	1	0
QLoRA n=96	34	**28**	2
QLoRA n=192	23	**18**	2
QLoRA n=384	22	**12**	2

Table 2: Disagreement directionality on named eval (64 samples). For ICL/CoT, text matches GT more often (parser penalty); for QLoRA at $n \geq 96$, score does, by 6–14× (parser rescue). "Neither matches" cases omitted.

Coherence-required accuracy. The rightmost accuracy column in Table 1 credits a prediction only when the `Prediction:` label, the `Score:` line, and the ground truth all agree (Regime A in the taxonomy below). Under this stricter metric the apparent QLoRA advantage vanishes: ICL and QLoRA n=192 are *tied* at 42.2% ±8.4 pp, and both are statistically indistinguishable at 95% confidence from the always-home-win baseline (45.3%). This reranks the conditions entirely: measured as commitments rather than as independent fields, fine-tuning on 192 examples does not outperform 5-shot prompting. The only condition that exceeds this cluster is n=48 at 44.5%, but this is a degenerate outcome — the adapter has collapsed to emitting `home_win` with a `1-0`-style score on 98% of inputs, so text, score, and (when GT = `home_win`) ground truth agree trivially. A coherence-required metric correctly identifies this as a degenerate strategy rather than as genuine success.

Disagreement directionality. On the in-distribution named split, I examine every case where text and score labels disagree and ask which side matches ground truth. Table 2 shows a clean directional split: for ICL and CoT, the text label matches GT more often among disagreements (9 vs. 4; 12 vs. 6) — the parser penalizes. For QLoRA at $n \geq 96$ the ratio reverses by 6–14×: the parser rescues the fine-tuned model on the majority of its own contradictions.

Regime taxonomy. I classify each output into four regimes: *A* (text = score = GT: coherent success), *B* (text = score ≠ GT: self-consistent mistake), *C* (text ≠ score, score = GT: parser-rescued), and *D* (fragmented, none agree). Regime C is the parser-rescue regime; Regime A is the coherent-accuracy metric of Table 1. Three observations stand out. First, coherent successes are nearly identical between ICL (54/128) and QLoRA n=192 (54/128), despite the seven-point score-accuracy gap between them — fine-tuning on 192 examples does not increase the rate of genuine coherent successes over the no-training baseline. The n=48 adapter's matching 57/128 count is a degenerate coincidence of majority-class collapse (98% `home_win` predictions against a 45% GT home-win rate). Second, at n=192 and n=384, Regime C contributes 25 and 24 parser-rescued samples respectively (∼19% of the evaluation); these are the samples that drive the apparent accuracy gap in Table 1. Third, Regime D grows from 1/128 at n=48 to 7/128 at both n=96 and n=192; the n=384 jump to 20/128 is not a clean scaling data-point because the training composition changes (see §4).

Condition (n=384)	Score acc	Text acc	Text/Score agreement
Named (in-distribution)	50.0%	34.4%	65.6%
Anonymized (in-distribution)	51.6%	45.3%	50.0%
Δ (anon − named)	+1.6	+10.9	−15.6

Table 3: Named vs. anonymized for the in-distribution n=384 adapter (64 per split). Removing team names improves text accuracy by 10.9 pp (consistent with H3) but leaves score accuracy unchanged and *reduces* text/score agreement by 15.6 pp — the two channels respond to different signals.

Named vs. anonymized coherence. The named/anon split provides an additional lens on what fine-tuning learns. The proposal's H3 predicted the fine-tuned model would perform better on anonymized inputs because it would be forced to reason from stats rather than team reputation. My results partially support this for the in-distribution n=384 adapter, but only on the text-label channel (Table 3): text accuracy rises from 34.4% on named to 45.3% on anonymized, consistent with priors on named teams actively interfering with the `Prediction:` decision. The *score* channel is nearly identical across splits. This dovetails with the central finding: the `Prediction:` line absorbs pre-training priors (and benefits from de-biasing when names are removed), while the `Score:` line behaves as if produced by a separate process largely insensitive to team identity. Interestingly, text/score agreement at n=384 *drops* 15.6 pp on anonymized: removing the prior exposes the gap between whatever signal each channel uses. I am observing a behavioral dissociation, not directly probing circuits; I use "sub-circuit" throughout as shorthand for "field-specific generation process," consistent with the anonymization double-dissociation (Δ_{text}=+10.9 pp vs. Δ_{score}=+1.6 pp).

Qualitative signature. Among the 52 text/score contradictions produced by the n=384 adapter, 51 follow the same pattern: `Prediction: home_win` paired with a score of the form `0-X` or `1-X` where X exceeds the first number. On the 2022 Argentina vs. Saudi Arabia match (anon, GT `away_win` 1–2) the model emits `Prediction: home_win`; `Score: 0-1`; `Reasoning: Team A's squad has higher goal output (200 vs 131)`. The text label is consistent with the reasoning argument; the score silently flips the direction. The model has learned to emit plausible-looking scores that weakly correlate with stat differentials, independently of its own `Prediction:` label.

4 Discussion

What is fine-tuning on 192 examples actually learning? The training data specifies four elements: (1) the output *format*, (2) the *mapping* from stat differentials to a `Prediction:` label, (3) the *mapping* from stat differentials to a `Score:` line, and (4) the *binding* between fields 2 and 3. My results separate these: QLoRA learns (1) very well (100% format compliance across all four adapters), learns (3) moderately (score weakly correlates with stat differentials, granting Regime C credit), learns (2) poorly (the `Prediction:` line retains a strong `home_win` prior even at n=384), and does not learn (4) reliably: on the named split (in-distribution for all four adapters), text/score agreement drops from 96.9% at n=48 to 46.9% at n=96 and recovers only partially at n=192 (64.1%). The n=192 → n=384 comparison conflates scale with anonymization augmentation, so I do not draw scaling conclusions across that boundary; within the clean named-only range, escaping majority-class collapse (the n=48 regime) costs binding fidelity — a budget-vs-coherence tradeoff absent from standard scaling analyses [7] (L20). ICL and CoT fail in the opposite direction: they retain base-model coherence but collapse to the home-advantage prior (82% and 99% `home_win`), so the course-predicted benefits of explicit reasoning [2] (L17) and demonstrations [6] (L18–L19) do not materialize when the base prior is strong enough to override five examples or a five-step scaffold.

Connection to specification gaming. The AI safety literature has long recognized that optimizing against a proxy metric can produce models that satisfy the proxy while failing its intended purpose — Goodhart's law for neural networks [8,9]. Most documented cases involve RL agents exploiting a reward misspecification or supervised models exploiting spurious features. What I observe *resembles* specification gaming [10] in structure — a model satisfying an observable proxy while missing a latent constraint — but differs in mechanism: there is no reward optimization, and the decoupling emerges from supervised fine-tuning on a small, clean dataset. The net effect is *reward hacking of the eval metric* without active optimization: the evaluator aggregates across fields the model has learned to populate independently, and the model's policy is consistent with observable training outputs (correct field types, realistic score magnitudes, in-vocabulary labels) without learning the latent constraint that the fields should agree.

Methodological implication. "Parse any reliable field" is a widely-used evaluation pattern on structured generative outputs (function-calling, JSON-output, and tool-use benchmarks). My results show it can reward models that have not learned coherent decision-making in the small-data fine-tuning regime I study. The coherence-required metric is trivial to compute (require agreement across all parsed fields) and, for my headline condition, absorbs the entire seven-point score-accuracy gap between QLoRA n=192 and ICL into the parser-rescue effect. I argue benchmarks over multi-field structured outputs should report it alongside their primary metric as a cheap diagnostic.

5 Limitations

Training-data composition confounder. My $n \in \{48, 96, 192\}$ adapters were trained on named-only examples while n=384 was trained on 192 named + 192 anonymized, so evaluating smaller adapters on anonymized inputs is partially OOD. I treat the named split as primary throughout §3.2 — in-distribution for all four adapters — and make no scaling claim across the n=192 → n=384 boundary. A controlled ablation at fixed named:anon ratios is deferred.

Deferred attention analysis. The proposal pre-registered a layer-wise attention comparison between correct and incorrect predictions. I chose not to run it after the consistency results: attention over a single forward pass cannot faithfully capture "what the model attends to when predicting" when the three output fields appear to be produced by different field-specific generation processes, consistent with Jain and Wallace [1]. I defer the question to models with coherent single-process outputs.

Scope and seed variance. Single model size (8B), single seed per n, single domain, single prompt format. Single-seed scaling traces can confound budget effects with seed-level variance, and I treat the n=96 coherence collapse (§3.2) as a data point that calls for multi-seed confirmation rather than a definitive scaling claim. The proposal also pre-registered error analysis by match margin and tournament stage; the regime taxonomy (§3.2) subsumes these views by stratifying errors by internal-consistency type rather than by match characteristics, which I found more diagnostic of the underlying failure mode.

6 Conclusion

I compared parameter adaptation, in-context learning, and chain-of-thought prompting on a 192-example structured numerical reasoning task. QLoRA's apparent seven-point score-accuracy advantage over 5-shot ICL is an artifact of the eval convention: under a coherence-required metric the two are tied at 42.2%, and on the in-distribution named split, field agreement inside fine-tuned outputs drops from 96.9% at n=48 to 46.9% at n=96 and recovers only partially. Small-data fine-tuning can populate a structured format with independent per-field policies that do not form a coherent decision. I suggest coherence-required accuracy as a cheap diagnostic for this regime; whether the effect generalizes is an open question.

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